RAMAKRISHNA MISSION VIDYAMANDIRA

(Residential Autonomous College affiliated to University of Calcutta)

B.A./B.Sc. SIXTH SEMESTER EXAMINATION, MAY 2023

THIRD YEAR [BATCH 2020-23]

COMPUTER SCIENCE [Honours]

Paper : DSE3

Date : 17/05/2023

Time : 11 am – 1 pm

Full Marks : 50

Answer **any five** questions of the following : [5×10]

1. (a) The following table gives dataset about Insured vehicles. Using Naïve Bayes classifier classify the new data (White, Sedan, Domestic).

Color	Type	Origin	Insured
Brown	Hatchback	Domestic	No
Brown	Hatchback	Imported	No
Brown	Sedan	Domestic	Yes
Orange	Hatchback	Domestic	Yes
Orange	Hatchback	Imported	Yes
Orange	Sedan	Imported	No
Orange	Hatchback	Imported	Yes
Orange	Sedan	Domestic	Yes
White	Sedan	Imported	No
White	Hatchback	Domestic	Yes

(b) Discuss the term: Gini index, Entropy, feature importance. (7+3)

2. (a) Perform the K-Means clustering step-by-step on the following dataset, considering K=2.

Products	Quantity	Price (K)
Laptop	3	9
Mobile	7	4
ipad	5	100
ipod	7	70

(b) What is Chi-Square, F_Claasif, Mutual_F_Classif? (7+3)

3. (a) Consider the following dataset where we have a data on the Systolic Blood Pressure (SBP) of two men and two women at different ages. Use Multiple Linear Regression to predict new SBP of new person. (Hints. Let SBP=99.2+0.42.Age+12.9.Gender). Assume that the both men and women have same slope. What will happen if both men and women have different slope.

SBP (mm Hg)	Age (years)	Gender
118	28	Woman
130	45	Man
122	34	Man
128	26	Woman

(b) What is Elbow method? (7+3)

4. (a) Write and explain the KNN algorithm for the following table.

Example	Shape	Size	Color	Surface	Thickness	Target Concept
1	Circular	Large	Light	Smooth	Thick	Malignant
2	Circular	Large	Light	Irregular	Thick	Malignant
3	Oval	Large	Dark	Smooth	Thin	Benign
4	Oval	Large	Light	Irregular	Thick	Malignant
5	Circular	Small	Light	Smooth	Thick	Benign

(b) How do you ensure you're not overfitting with a model? (7+3)

5. (a) Write and explain the Decision Tree (step-by-step) for the following table.

Example	Sky	Air Temp	Humidity	Wind	Water	Forecast	Enjoy Sport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Warm	Change	Yes

(b) Explain different kinds of cross validation techniques? (7+3)

6. (a) Explain the Principle Component Analysis (PCA) algorithm with an example.

(b) How would you handle an imbalanced dataset? (7+3)

7. (a) Perform a suitable ML algorithm on the following dataset for find out the infection status for 35.2 CRP(mg/L) and 40 Temp (C).

Infection	CRP(mg/L)	Temp (C)
Viral	40.0	36.0
Viral	11.1	37.2
Viral	30.0	36.5
Bacterial	42.0	37.6
Bacterial	31.1	42.2
Bacterial	50.0	38.5

(b) Explain the different steps of Gradient descent along with proper derivation. (7+3)

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